

STUDENT LEARNING OS

Zynth

Most study apps show you content.

Zynth shows you the truth about what you actually know — and rebuilds your plan around it, live.

● UNTOUCHED

● ENGAGED

● PROVEN

THE PROBLEM

Students confuse having studied something with actually knowing it.

"I studied X"

You were exposed to it. Nothing more.

"I understand X"

You feel fluent. Feelings are unreliable here.

"Here is proof"

Something external tested you, and you passed.

Re-reading feels smoother every pass, and the smoothness gets mistaken for understanding. It is the most common way students lose a term.

WHY EDTECH DOESN'T FIX IT

Everything either teaches, or tests. Almost nothing diagnoses.

CONTENT-FIRST

Tools that teach

Will happily explain a concept for the tenth time. No idea whether it landed, no memory of the last nine attempts.

QUIZ-FIRST

Tools that test

Give you a score. Don't diagnose *why* you got it wrong, and don't change what you do tomorrow.

Zynth exists in the gap: diagnose continuously, and re-plan continuously.

THE INSIGHT

Colour is evidence, not effort.

● RED

Untouched. Re-reading the chapter nine times does not move it.

● AMBER

You engaged with it. Zynth believes you understand — but has no evidence.

● GREEN

You passed a quiz. The only route here.

Amber is the state most products don't have. It's the honest name for "you did the work, but nobody checked."

THE CORE MECHANIC

Three legal transitions.
There is no red → green.

red —engage→ amber

amber —quiz passed ≥ 70 → green

green —failed retest→ amber

Enforced twice: one service owns every status write, and a SQLite trigger rejects illegal transitions at the storage engine. Even a raw UPDATE can't cheat it. That makes it an invariant, not validation.

THE PRODUCT

Your whole syllabus as one living 3D map.

Every concept is a node. Every module in the product reads from this graph and writes back to it — so the map is the app's state, not a picture of it.

Clustered by subject

Prerequisite edges

Live over WebSockets

Click a node → a real screen

Multiple graphs as tabs

Three steps from stuck to proven.

STEP 01

Intuition

One slider to drag, then a prediction you must commit to before anything is revealed. Guess wrong and your curve is drawn against reality.



STEP 02

Explain

A tutor that already holds your file, and teaches the exact objective the quiz is built from.



STEP 03

Quiz

The only route to green. Free-response graded by a different model than the one that wrote it.

Everything the quiz can ask was taught on the previous screen — true by construction, not by luck.

THE DIFFERENTIATOR

It rewires your map around how **you** fail.

Paste in a term's worth of wrong answers. The Autopsy Board finds the single misconception underneath all of them — then draws new edges between the concepts that keep failing together.

Not "you're bad at related rates."

Instead: **you drop the negative when the inner function is decreasing.**

WHY IT STAYS HONEST

Green decays.

A proven concept that fails a retest drops straight back to amber. Zynth will withdraw a claim it made about you.

Mastery isn't a badge you keep.
It's a claim the graph **keeps re-checking**.

No XP, no levels, no streaks-for-logging-in. The only streak counts nodes that stayed green through retests.

UNDER THE HOOD

AI generates and diagnoses. Deterministic code decides.

Gemini

Visuals, quizzes, the tutor, mistake clustering, planning

Groq · Llama 3.3 70B

Grades free response — never marks its own homework

No model can

Move a node to green, order prerequisites, or summon a Co-Pilot card

React · Three.js · Node · SQLite (WAL) · Socket.io · zynth-delta.vercel.app ·
github.com/AdamACE9/zynth